Factor-Graph Aeromagnetic Compensation for Airborne Magnetic-Anomaly Navigation

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***Abstract*—Airborne magnetic-anomaly navigation bounds inertial drift in GNSS-denied flight by matching a scalar magnetometer against a magnetic-anomaly map. Its central obstacle is aeromagnetic interference: the aircraft's own field must be compensated online when no calibration flight is available (the cold-start regime). A recent approach augments an extended Kalman filter (EKF) with online Tolles–Lawson (TL) coefficients and a residual neural network. We show that a factor-graph formulation reaches the same accuracy without a neural network by carrying the TL coefficients as graph variables of a fixed-lag sliding-window smoother, so the compensation adapts along the flight. On the primary evaluation line of the SGL 2020 dataset the sliding-window smoother matches or beats a reproduced online EKF+TL+NN and the values published for it. Across five long survey lines from three flights and two maps, it improves on a causal online-TL EKF on 8 of 9 cold-start cases and, importantly, remains bounded where the causal filter diverges. We also give an exact linear condition for the joint observability of compensation and navigation, from which an exogeneity design rule for the compensation features follows.**

***Index Terms*—Magnetic-anomaly navigation, aeromagnetic compensation, Tolles–Lawson, factor-graph optimization, fixed-lag smoothing, GNSS-denied navigation.**

NOMENCLATURE

$\mathbf{x}_t$	INS error state at epoch t (position, velocity, attitude, biases, FOGM).
$\boldsymbol{\beta}_t$	Online Tolles–Lawson compensation coefficients.
$\boldsymbol{\chi}_t$	Joint state $(\mathbf{x}_t, \boldsymbol{\beta}_t)$.
$\boldsymbol{\Phi}_t, \mathbf{Q}_t$	Pinson state-transition matrix and process covariance.
z_t, η_t	Scalar magnetometer measurement and its noise, variance R.
$h(\cdot)$	Anomaly-map interpolation (with IGRF core) as a function of position.
$\mathbf{g}_t$	Map gradient $\partial h/\partial \mathbf{p}$ at epoch t.
$\mathbf{A}_t$	Tolles–Lawson basis row from the three-axis fluxgate.
c_t	Aircraft magnetic interference, $c_t = \mathbf{A}_t^\top \boldsymbol{\beta}_t$.
L_w, L_o	Sliding-window length and overlap (look-ahead).
ρ, δ	Robust kernel and its Huber threshold.
$\mathbf{G}, \boldsymbol{\Psi}$	Stacked map-gradient and TL-regressor matrices over a window.

I. INTRODUCTION

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Reproducible code and experiments are provided in the MagNav.jl research branch `claude/fgo-problem-research-bllgbr`.

INERTIAL navigation systems (INS) are the backbone of airborne navigation, but their position error grows without bound as accelerometer and gyroscope errors are integrated over time, so an INS must be aided by an absolute position reference [1], [2]. The satellite-based reference of choice, GNSS, is unavailable or untrustworthy in the contested and signal-degraded environments that increasingly define aerospace and defense operations. This has renewed interest in passive, self-contained references that draw on natural geophysical signatures of the Earth. Among these, the crustal magnetic-anomaly field is attractive: it is globally available, temporally stable over the mission horizon, difficult to deny or spoof, and observable with a lightweight scalar magnetometer [3], [4].

Magnetic-anomaly navigation (MagNav) estimates aircraft position by matching the measured scalar magnetic field against a surveyed anomaly map, yielding an absolute fix that bounds inertial drift [4]. The public benchmark that has catalyzed recent work is the Signal Enhancement for Magnetic Navigation (SGL 2020) challenge dataset, comprising flight-grade INS, multiple onboard magnetometers, and high-resolution anomaly maps [5]. That dataset also exposes the central obstacle: the dominant error source is the aircraft itself, not the map. The platform's own permanent and induced magnetization and its eddy-current fields perturb the scalar measurement by hundreds of nanotesla, comparable to or larger than the anomaly signal being matched, so the aircraft field must be compensated before map matching can recover position.

Aeromagnetic compensation is classically handled by the Tolles–Lawson (TL) model [6], which regresses the interference onto sixteen attitude-dependent terms—permanent, induced, and eddy-current—formed from a three-axis fluxgate magnetometer. The coefficients are conventionally identified on a dedicated high-altitude calibration flight of prescribed maneuvers, where the anomaly signal is negligible and the interference dominates. Many operational scenarios, however, admit no such calibration flight: the coefficients drift with temperature and configuration, and a survey may begin with no prior calibration at all. In this cold-start regime the compensation coefficients must be learned online and jointly with navigation, from the same measurements used to fix position—coupling that makes both estimation and its analysis substantially harder than the calibrated case.

Recursive Bayesian filtering is the standard machinery for map-matching navigation, from the linear Kalman filter [7] to nonlinear variants. Geophysical map matching has a long lineage: terrain-contour matching for cruise missiles [8] and nonlinear Kalman filtering for terrain-aided navigation [9] established the paradigm, and grid- and particle-based Bayesian estimators later became standard where the measurement-to-position map is strongly nonlinear or multimodal [10], [11], [12], [13]. For MagNav specifically, Canciani and Raquet combined a calibrated scalar magnetometer with an extended Kalman filter (EKF) and characterized the achievable accuracy on flight data [3], [4], later demonstrating airborne magnetic navigation with online calibration [14]. Beyond the EKF, unscented filtering [15], invariant formulations that respect the state manifold [16], and mixture/particle representations [17] improve consistency under nonlinearity. *However*, these estimators are causal: they commit each state online with no access to future measurements, and the position-centric ones carry no compensation state of their own. When a large uncompensated cold-start residual enters, a causal filter has no mechanism to attribute the mismatch to the aircraft field rather than to position, and the estimate can diverge.

A second line of work makes the compensation itself adaptive. Classical Tolles–Lawson (TL) identification is a batch regression on a calibration flight [6], [18], [19]; to remove that requirement the TL regression can be embedded in the estimator and its coefficients tracked online as additional states. Nonlinear extensions replace or augment the linear basis with learned components: neural-network compensators date back three decades [20], model-stability and robustness of the regression have been studied in gradiometry [21], and machine-learning-enhanced calibration has recently been applied to the SGL benchmark [22]. Most relevant here, a residual neural network (NN) has been added on top of an online TL model, carried inside an EKF and trained online, to capture interference the linear basis misses [23]; this reaches good cold-start accuracy on the SGL 2020 cabin magnetometers and is, to our knowledge, the strongest published cold-start result on that benchmark. *Nevertheless*, the network requires a carefully engineered online-training design to remain stable, adds opaque parameters whose behavior is hard to certify, and—being embedded in an EKF—inherits the same causal, commit-online limitation as the filters above.

A third body of work, largely from robotics, replaces filtering with factor-graph optimization: the estimation problem is a graph of variables and probabilistic constraints [24] whose maximum-a-posteriori (MAP) solution is found by sparse nonlinear least squares [25], [26], the same structure underlying bundle adjustment and graph SLAM [27], [28], [29]. Incremental and square-root smoothers—square-root SAM [30], iSAM [31], iSAM2 [32], and exactly-sparse delayed-state filters [33]—make the solution efficient enough for online use, and fixed-lag sliding-window optimization [34], [35] bounds its cost. On-manifold inertial preintegration [36], factor-graph inertial fusion [37], and modern visual–inertial estimators [38] have brought the approach firmly into navigation. *Despite these advances*, factor-graph estimation in navigation has been developed almost entirely for visual–inertial odometry and simultaneous localization and mapping, where the estimated quantities are poses and landmarks; it has not been brought to bear on *joint* online aeromagnetic compensation and map matching, and the observability of estimating a compensation model jointly with position from map matching has not been characterized.

In this work we cast online Tolles–Lawson compensation as a set of time-varying variables in a factor graph and solve the joint compensation-and-navigation problem as a fixed-lag sliding-window smoother. The per-window relinearization supplies the time-varying compensation that the competing method obtains from an online network, and deferring commitment over the window removes the causal filter's failure mode. We are explicit about scope: for the linear–Gaussian INS error model the batch MAP estimate on the factor-graph chain equals the Rauch–Tung–Striebel (RTS) smoother [39], so smoothing per se is not the contribution; the joint windowed estimation of a time-varying compensation is. The paper makes the following contributions:

1) **NN-free online compensation:** we formulate online TL compensation as factor-graph variables in a fixed-lag sliding-window smoother, an alternative to online EKF+TL+NN that needs no neural network (Section III).
2) **Observability condition:** we prove an exact linear condition for the joint observability of compensation and navigation and derive an exogeneity design rule for the compensation features (Section IV).
3) **Reproduced baseline and breadth study:** we reproduce—rather than cite—the online EKF+TL+NN result and evaluate both estimators on five long survey lines from three flights and two maps (Section VI).
4) **Sensor-error factors:** we cast physically motivated scalar-magnetometer error terms as graph variables that a position-only estimator cannot represent (Section III).

The remainder of this paper is organized as follows. Section II states the INS error model, the measurement, and the batch MAP problem. Section III develops the sliding-window smoother and the compensation and sensor-error factors. Section IV establishes the observability condition. Section V describes the reproduced EKF+TL+NN baseline, and Section VI reports the experiments. Section VII discusses the results and their limitations, and Section VIII concludes.

II. Problem Formulation

We first fix notation and state the estimation problem. Throughout, boldface lowercase denotes vectors and boldface uppercase matrices; $\mathcal{N}(\cdot,\cdot)$ is the Gaussian density; and $\|\mathbf{v}\|^2_{\mathbf{M}} = \mathbf{v}^\top \mathbf{M} \mathbf{v}$ is the squared Mahalanobis norm.

A. INS error state and dynamics

We adopt the standard Pinson error model of a strapdown INS [2], [1], [40]. The error state

$$\mathbf{x} = \begin{bmatrix} \delta\mathbf{p}^\top \ \delta\mathbf{v}^\top \ \boldsymbol{\psi}^\top \ \mathbf{b}_a^\top \ \mathbf{b}_g^\top \ S \end{bmatrix}^\top \in \mathbb{R}^n \tag{1}$$

stacks position error $\delta\mathbf{p}$, velocity error $\delta\mathbf{v}$, attitude error $\boldsymbol{\psi}$, accelerometer and gyroscope biases $\mathbf{b}_a, \mathbf{b}_g$, and a scalar first-order Gauss–Markov (FOGM) state S that absorbs residual magnetic disturbance and map error. The discretized, linearized dynamics are

$$\mathbf{x}_{t+1} = \boldsymbol{\Phi}_t \mathbf{x}_t + \mathbf{w}_t, \qquad \mathbf{w}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}_t), \tag{2}$$

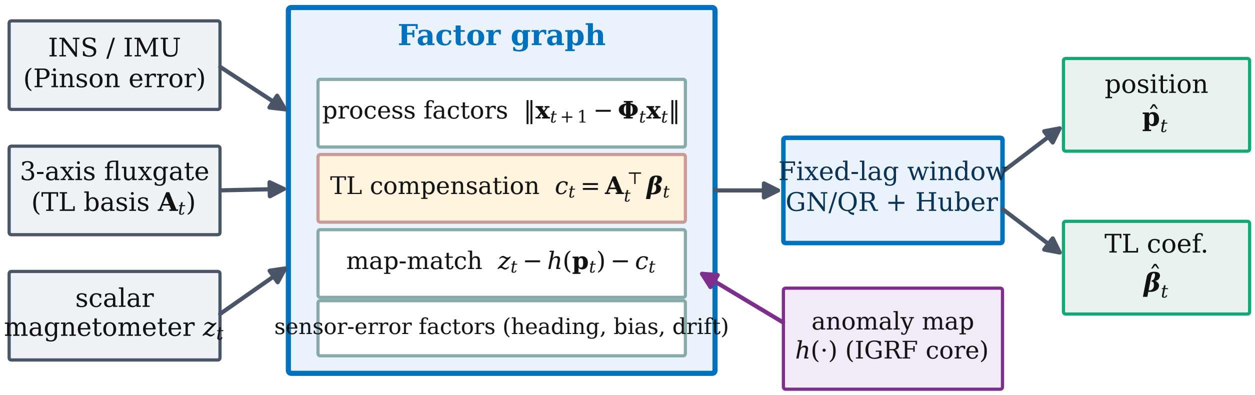

Fig. 1. Proposed estimator. The INS error state, the three-axis fluxgate (Tolles–Lawson basis), and the scalar magnetometer feed a factor graph whose variables are the navigation error, the online TL coefficients, and optional sensor-error states. A fixed-lag window solves the graph by robust Gauss–Newton/QR and outputs position and compensation; the anomaly map enters through the map-match factor.

where $\mathbf{\Phi}_t$ is the Pinson state-transition matrix and $\mathbf{Q}_t$ the process covariance; the FOGM block of $\mathbf{\Phi}_t$ is $e^{-\Delta t/\tau}$ with correlation time τ.

B. *Scalar magnetometer measurement*

The scalar magnetometer reads

$$z_t = h(\mathbf{p}_t) + c_t + \eta_t, \qquad \eta_t \sim \mathcal{N}(0, R), \tag{3}$$

where $h(\cdot)$ interpolates the crustal anomaly map (with the IGRF core field restored) at the aircraft position $\mathbf{p_t} = \bar{\mathbf{p}}_t + \delta\mathbf{p}_t$ and c_t is the aircraft interference. Linearizing about the nominal position,

$$z_t \approx h(\bar{\mathbf{p}}_t) + \mathbf{g}_t^\top \delta\mathbf{p}_t + c_t + \eta_t, \qquad \mathbf{g}_t = \left.\frac{\partial h}{\partial \mathbf{p}}\right|_{\bar{\mathbf{p}}_t}, \tag{4}$$

so the map gradient $\mathbf{g}_t$ is the *sole* source of horizontal position information: where $\mathbf{g}_t \approx \mathbf{0}$ (a locally flat map) position is momentarily unobservable, and everywhere the interference c_t competes with the $\mathbf{g}_t^\top \delta\mathbf{p}_t$ term for the same measurement.

C. *Tolles–Lawson interference model*

Classical aeromagnetic compensation writes the interference as a linear combination of attitude-dependent basis terms [6], [18], [19],

$$c_t = \mathbf{A}_t^\top \boldsymbol{\beta}_t, \tag{5}$$

where $\mathbf{A}_t \in \mathbb{R}^m$ is the TL basis row built from the three-axis fluxgate. With direction cosines $\hat{\mathbf{b}}_t = [\,u\ v\ w\,]^\top$ of the Earth field in body axes and total field magnitude B_t, the permanent (3), induced (6), and eddy-current (9) groups are

$$\mathbf{A}_t = \big[\,\underbrace{u, v, w}_{\text{perm.}},\ \underbrace{B_t u^2, \ldots, B_t w^2}_{\text{induced}},\ \underbrace{B_t u\dot{u}, \ldots, B_t w\dot{w}}_{\text{eddy}}\,\big]^\top, \tag{6}$$

optionally augmented by a bias term, giving $m \le 19$ coefficients $\boldsymbol{\beta}_t$. In the calibrated setting $\boldsymbol{\beta}$ is fit once and frozen; in the cold-start setting it is unknown at takeoff and drifts slowly, so we treat it as a time-varying state with a random-walk prior $\boldsymbol{\beta}_{t+1} = \boldsymbol{\beta}_t + \mathbf{w}_t^\beta$, $\mathbf{w}_t^\beta \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}^\beta)$.

D. *Batch maximum-a-posteriori estimate*

Collecting the navigation and compensation variables into $\boldsymbol{\chi}_t = (\mathbf{x}_t, \boldsymbol{\beta}_t)$ over a window of N epochs, the MAP estimate minimizes

$$\begin{aligned} J(\boldsymbol{\chi}) = \|\mathbf{x}_1 - \mathbf{x}_0\|^2_{\mathbf{P}_0^{-1}} + \sum_{t=1}^{N-1} \|\mathbf{x}_{t+1} - \mathbf{\Phi}_t\mathbf{x}_t\|^2_{\mathbf{Q}_t^{-1}} \\ + \sum_{t=1}^{N-1} \|\boldsymbol{\beta}_{t+1} - \boldsymbol{\beta}_t\|^2_{(\mathbf{Q}^\beta)^{-1}} + \sum_{t=1}^{N} \rho\left(\frac{r_t(\boldsymbol{\chi}_t)}{\sqrt{R}}\right), \end{aligned} \tag{7}$$

with measurement residual $r_t(\boldsymbol{\chi}_t) = z_t - h(\mathbf{p}_t) - \mathbf{A}_t^\top \boldsymbol{\beta}_t - S_t$ and ρ a possibly robust kernel. We make the standard modeling assumptions of aided inertial navigation [1], [17].

Assumption 1: The process and measurement noises $\mathbf{w}_t, \mathbf{w}_t^\beta, \eta_t$ are zero-mean, white, mutually independent, and Gaussian; the linearizations (2) and (4) are valid over a window; and the prior $(\hat{\boldsymbol{\chi}}_0, \mathbf{P}_0)$ is Gaussian.

Assumption 1 is the same one under which the EKF and RTS smoother are derived, so any performance difference between the estimators reflects their *structure*, not their noise models. Under it the batch estimate coincides with the classical smoother.

Lemma 1: Under Assumption 1, with $\rho(u) = \frac{1}{2}u^2$ and $\boldsymbol{\beta}$ held fixed, the minimizer of (7) equals the Rauch–Tung–Striebel (RTS) smoother estimate over the window.

Proof: With ρ quadratic and $\boldsymbol{\beta}$ fixed, (7) is a positive-definite quadratic in the Gaussian chain $\{\mathbf{x}_t\}$, so its minimizer is the posterior mean $\mathbb{E}[\mathbf{x}_{1:N} \mid z_{1:N}]$. For a linear–Gaussian state-space model that conditional mean is exactly what the forward Kalman recursion followed by the backward RTS recursion computes [39]; the two therefore coincide. ■

Remark 1: Lemma 1 is why we do not claim smoothing itself as a contribution: for a static, known compensation the batch graph and a filter–smoother pass are equivalent. The contribution is what the graph makes easy that a filter does not—jointly estimating the *time-varying* $\boldsymbol{\beta}$, adding the non-Gaussian robust kernel of (10), and appending the sensor-error variables of Section III—together with the fixed-lag windowing that keeps it causal-latency bounded.

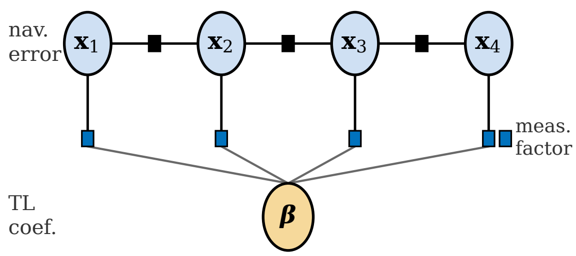

Fig. 2. Factor graph. The Tolles–Lawson coefficient variable $\boldsymbol{\beta}$ (shared, bottom) enters every measurement factor and is estimated jointly with the navigation-error chain $\mathbf{x}_1, \ldots, \mathbf{x}_N$.

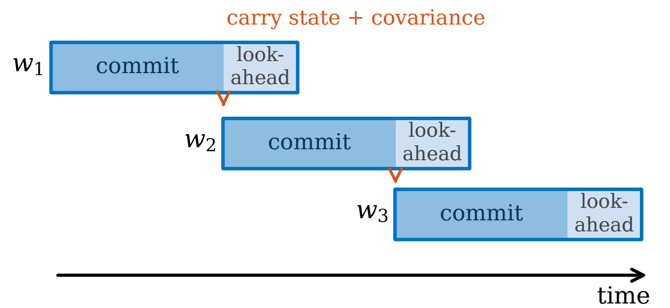

Fig. 3. Fixed-lag sliding window. Each window commits its leading stride and passes the full smoothed state and covariance forward as the prior for the next window; the trailing overlap provides bounded smoother look-ahead.

III. Method

The proposed estimator (Fig. 1) represents navigation, online compensation, and optional sensor-error states as variables of a single factor graph, and solves it over a fixed-lag sliding window by robust Gauss–Newton. This section develops the factors, the windowing, the robust solver, and the sensor-error terms.

A. Online TL coefficients as graph variables

We append the compensation coefficients $\boldsymbol{\beta}$ to the state, so the measurement factor of (3) becomes, in residual form,

$$r_t(\boldsymbol{\chi}_t) = z_t - h(\mathbf{p}_t) - \mathbf{A}_t^\top \boldsymbol{\beta}_t - S_t, \tag{8}$$

with Gaussian information R^{-1}, and its Jacobian row is

$$\mathbf{H}_t = \Big[\, \underbrace{\mathbf{g}_t^\top}_{\delta\mathbf{p}} \;\; \underbrace{\mathbf{0}}_{\delta\mathbf{v},\boldsymbol{\psi},\mathbf{b}} \;\; \underbrace{\mathbf{A}_t^\top}_{\boldsymbol{\beta}} \;\; \underbrace{1}_{S} \,\Big]. \tag{9}$$

The coefficient block is shared by every measurement factor in the window (Fig. 2), so the compensation is identified from the whole trajectory rather than instantaneously. Because the batch solution uses future as well as past measurements, early navigation states benefit from calibration information that only becomes observable after later maneuvers—the property a causal filter lacks. A random-walk factor $\|\boldsymbol{\beta}_{t+1} - \boldsymbol{\beta}_t\|^2_{(\mathbf{Q}^\beta)^{-1}}$ links consecutive coefficient blocks and sets how quickly the compensation may adapt; $\mathbf{Q}^\beta \to \mathbf{0}$ recovers a single static $\boldsymbol{\beta}$.

B. Fixed-lag sliding window

A single batch over a long line forces one static $\boldsymbol{\beta}$, which underfits time-varying interference; a per-epoch filter, conversely, cannot use future data. We take the middle path and process the flight in overlapping windows of length L_w with overlap L_o, each a batch factor graph, in the manner of fixed-lag and incremental smoothing [32], [34], [35]. Window k spans epochs $[s_k, s_k + L_w)$; it is solved as a batch, commits only its leading stride of $L_w - L_o$ epochs, and carries the full smoothed state and covariance of the last committed epoch forward as a Gaussian prior for window $k{+}1$ (Fig. 3). The trailing overlap L_o acts as bounded smoother look-ahead, so each committed estimate has seen L_o epochs of future data. Carrying the full covariance—not just the mean—prevents the double-counting of information that a naive re-linearization would incur, and keeps navigation continuous across window boundaries. Algorithm 1 summarizes the procedure; its per-window cost is constant in the line length, giving overall linear complexity.

Algorithm 1 Fixed-lag sliding-window smoother

1: **Input:** measurements $\{z_t\}$, TL rows $\{\mathbf{A}_t\}$, map h, window L_w, overlap L_o, prior $(\hat{\boldsymbol{\chi}}_0, \mathbf{P}_0)$
2: $s \leftarrow 1$
3: **while** $s \le T$ **do**
4: build factor graph on epochs $[s, \min(s{+}L_w, T))$ with prior $(\hat{\boldsymbol{\chi}}_{s-1}, \mathbf{P}_{s-1})$
5: **repeat** ▷ robust Gauss–Newton
6: linearize residuals (8); form weights w_t from ρ (11)
7: solve normal equations by sparse QR (square-root SAM) [30]
8: update $\boldsymbol{\chi} \leftarrow \boldsymbol{\chi} + \Delta\boldsymbol{\chi}$
9: **until** converged
10: commit epochs $[s, s{+}L_w{-}L_o)$; recover $(\hat{\boldsymbol{\chi}}, \mathbf{P})$ at epoch $s{+}L_w{-}L_o{-}1$
11: $s \leftarrow s + (L_w - L_o)$
12: **end while**
13: **Output:** committed trajectory $\{\hat{\mathbf{p}}_t, \hat{\boldsymbol{\beta}}_t\}$

C. Window design and initialization

Two parameters govern the window: the length L_w and the overlap L_o. L_w sets how much data each compensation estimate sees and thus the effective adaptation time; too short and $\boldsymbol{\beta}$ is under-determined, too long and it cannot track drift. L_o sets the look-ahead and hence the commit latency: a committed state has seen L_o epochs of future data, and the estimator lags real time by L_o. In the experiments we use $L_w = 5\,\text{min}$ and $L_o = 2\,\text{min}$; the $2\,\text{min}$ window in Table II shows that halving L_w degrades the long-line result, consistent with under-determination. The first window is seeded with a diffuse prior on $\boldsymbol{\beta}$ (large $\mathbf{P}_0$ block) so the cold start is not biased toward zero interference, and the FOGM disturbance state absorbs the initial mismatch while the coefficients converge; subsequent windows inherit the tight carried covariance and converge in one or two Gauss–Newton iterations. This warm-start is why the per-window iteration count stays low despite the nonlinearity of the map term.

D. Robust kernel and reweighting

Map error, transient interference, and dead-zone measurements produce heavy-tailed residuals that a quadratic cost over-weights. We use a Huber kernel [41]

$$\rho(u) = \begin{cases} \frac{1}{2}u^2, & |u| \le \delta, \\ \delta\big(|u| - \frac{1}{2}\delta\big), & |u| > \delta, \end{cases} \tag{10}$$

minimized by iteratively reweighted least squares (IRLS): each Gauss–Newton iteration reweights measurement factor t by

$$w_t = \min\big(1,\ \delta/|u_t|\big), \qquad u_t = r_t/\sqrt{R}, \tag{11}$$

so gross outliers are down-weighted toward an absolute-value penalty while inliers keep the efficient quadratic cost. Alternatives such as switchable constraints [42] and dynamic covariance scaling [43] fit the same graph interface and could replace (10) unchanged.

E. Sparse solver

Each Gauss–Newton iteration solves a sparse linear system whose Jacobian has the block-bidiagonal-plus-shared-column structure of Fig. 2. We factor it by column-pivoted QR on the square-root information matrix (square-root SAM [30], [25]), which is numerically stable and exploits the chain sparsity; an iterated RTS pass [39] gives the identical estimate for the Gaussian case and we verify the two agree to sub-metre level in Section VI.

F. Physical sensor-error factors

The graph representation admits error states that a position-only grid or point-mass estimator cannot carry. Beyond TL we optionally augment $\boldsymbol{\chi}$ with four physically motivated scalar-magnetometer terms, each appearing additively in the residual (8):

1) *Heading error*, modeled as a truncated Fourier series in the sensor–field angle θ_t, $b_t^{\mathrm{hdg}} = \sum_{k=1}^{K} \big(a_k \cos k\theta_t + d_k \sin k\theta_t\big)$, with coefficients $\{a_k, d_k\}$ as graph variables;
2) *Hard-iron bias* b^{hi}, a constant offset variable;
3) *Linear drift* $b_t^{\mathrm{dr}} = \gamma_0 + \gamma_1 t$, capturing slow electronics drift;
4) *Dead-zone weight*, a heteroscedastic measurement information $R_t^{-1} = R^{-1}\,\omega(z_t)$ that de-weights samples in a sensor dead band.

The augmented residual is $r_t = z_t - h(\mathbf{p}_t) - \mathbf{A}_t^\top \boldsymbol{\beta}_t - S_t - b_t^{\mathrm{hdg}} - b^{\mathrm{hi}} - b_t^{\mathrm{dr}}$. Each term is exogenous to position (Section IV), so it can be identified without stealing map information; their observability is limited only by how much the flight excites the corresponding regressor (e.g. the heading harmonics need attitude variation).

IV. Observability of Joint Compensation and Navigation

Whether position can be recovered while the compensation is estimated jointly is a property of the measurement geometry. Over a window let $\mathbf{G}$ have rows $\mathbf{g}_t^\top$ (the map-gradient/position sensitivity) and let $\boldsymbol{\Psi}$ have rows $\mathbf{A}_t^\top$ (the TL regressor). To isolate the position–compensation coupling, restrict (3) to the $(\delta\mathbf{p}, \delta\boldsymbol{\beta})$ subspace, giving the window model $\mathbf{z} = \mathbf{G}\,\delta\mathbf{p} + \boldsymbol{\Psi}\,\delta\boldsymbol{\beta} + \boldsymbol{\eta}$.

Proposition 1: A position perturbation $\delta\mathbf{p}$ is indistinguishable from a compensation adjustment—hence unobservable from map matching over the window—if and only if $\mathbf{G}\,\delta\mathbf{p} \in \mathrm{range}(\boldsymbol{\Psi})$. Consequently the joint state $(\delta\mathbf{p}, \delta\boldsymbol{\beta})$ is observable if and only if $\mathrm{range}(\mathbf{G}) \cap \mathrm{range}(\boldsymbol{\Psi}) = \{\mathbf{0}\}$.

Proof: Two joint states $(\delta\mathbf{p}, \delta\boldsymbol{\beta})$ and $(\delta\mathbf{p}', \delta\boldsymbol{\beta}')$ produce identical noise-free predictions iff $\mathbf{G}(\delta\mathbf{p} - \delta\mathbf{p}') = \boldsymbol{\Psi}(\delta\boldsymbol{\beta}' - \delta\boldsymbol{\beta})$, i.e. iff $\mathbf{G}(\delta\mathbf{p} - \delta\mathbf{p}') \in \mathrm{range}(\boldsymbol{\Psi})$. A nonzero position difference with this property exists iff $\mathrm{range}(\mathbf{G}) \cap \mathrm{range}(\boldsymbol{\Psi}) \neq \{\mathbf{0}\}$, which is exactly the negation of the stated condition; equivalently the stacked Jacobian $[\,\mathbf{G} \quad \boldsymbol{\Psi}\,]$ is column-rank-deficient. ■

Corollary 1 (Exogeneity): If $\mathbf{A}_t$ is a function of aircraft attitude/fluxgate alone—exogenous to position—then $\mathrm{range}(\boldsymbol{\Psi})$ is generically transverse to the map-gradient subspace $\mathrm{range}(\mathbf{G})$ and the joint problem is observable at any basis dimension. If instead a compensation feature contains the map value $h(\mathbf{p})$ (for example the scalar magnetometer itself), its column aligns with $\mathrm{range}(\mathbf{G})$, forcing the intersection nonzero and collapsing observability.

Proposition 1 is exact but non-constructive: it does not yield a reliable scalar screening statistic, and in our experiments correlation-based proxies for the intersection (in-sample basis R^2, out-of-sample prediction, and gradient-alignment indices) did not generalize across estimators or lines. The actionable consequence is Corollary 1: drive the compensation from exogenous features, which every estimator in Section VI does.

Remark 2: The condition is geometric, not statistical: it depends on the ranges of $\mathbf{G}$ and $\boldsymbol{\Psi}$ over the window, hence on the flight path and the local map gradient, not on noise levels. It explains two failure modes seen in practice. First, over a locally flat map $\mathbf{G} \approx \mathbf{0}$ and *no* compensation can be separated from position—a map-information floor that bounds every estimator, and that we observe on the shortest line in Section VI. Second, an *endogenous* feature such as the scalar magnetometer reading itself makes a $\boldsymbol{\Psi}$ column collinear with $\mathbf{G}$, so the compensator silently absorbs the map signal; excluding such features (Section V) is necessary for the baseline to converge.

V. Reproduced Online EKF+TL+NN Baseline

To compare against [23] on equal footing we *reproduce*, rather than cite, the online EKF+TL+NN cold start. The filter augments the Pinson error state with the online TL coefficients $\boldsymbol{\beta}$ and the weights of a small feed-forward network, all propagated as EKF states and updated online against (3). Naively initialized, the reproduction diverged; three fixes, each addressing a distinct and independently diagnosed failure, were required.

(i) Bias-free network with DC initialization. The map returns the anomaly only, whereas the raw scalar reading includes a large constant offset. Left in the regression target, this offset drove the network output bias to $\sim 5 \times 10^4$ nT and the filter to `NaN`. We remove the mean by a short initial window, initializing the network output bias to the median measured-minus-predicted offset and scaling the target by its standard deviation.

(ii) Exogenous network inputs only. Using the scalar magnetometer as a network input creates exactly the endogenous feature of the Remark above: the network learns to reproduce the map and horizontal position collapses (we observed $>$ 5 km error with a small measurement residual). Restricting the inputs to the attitude-derived TL basis restores observability, in direct agreement with Corollary 1.

(iii) De-trusting the map through the transient. During the cold-start transient the compensation is grossly wrong, so the map residual is unreliable; we inflate the measurement variance and the coefficient process noise for the warm-up, then relax them. With these three fixes the filter lands inside the band published for it [23] (Table II) and serves as our baseline.

VI. Experiments and Results

A. Dataset, maps, and protocol

All experiments use the public SGL 2020 challenge dataset [5]: a Cessna 208B instrumented with a flight-grade INS, several onboard scalar magnetometers (a compensated tail-stinger sensor and uncompensated cabin sensors Mag 4 and Mag 5), a three-axis fluxgate, and reference truth. Map matching uses the Eastern ("E") and Renfrew ("R") high-resolution anomaly grids, upward-continued to flight altitude with the IGRF core field restored. The metric is horizontal distance-RMS (DRMS) of position error after a $10\,\mathrm{min}$ warm-up. *Cold start* means the TL coefficients are initialized to zero with no calibration flight. For every comparison the EKF and FGO estimators use identical dynamics, map interpolation, and noise settings, so that only the estimator *structure* differs. Unless noted, the window smoother uses $L_w = 5\,\mathrm{min}$, $L_o = 2\,\mathrm{min}$, and the Huber kernel (10). All numbers are produced by continuous-integration scripts on the code branch and are reproducible from it.

B. Batch estimator on real data

We first validate the estimator on line 1003.02 (Flt1003, $63\,\mathrm{min}$) with the compensated stinger magnetometer, where the interference is already removed and the comparison isolates the estimator. The batch factor graph improves horizontal DRMS from the EKF's $28.3\,\mathrm{m}$ to $14.0\,\mathrm{m}$ (Huber kernel), against $114.6\,\mathrm{m}$ free-inertial—a $51\,\%$ reduction over the EKF. Fig. 4 shows the anomaly map, the flight line, and a mid-flight zoom in which the free-inertial track visibly departs from truth while the EKF and FGO estimates follow it. Fig. 5 plots the horizontal error over the line: the FGO estimate (blue) stays below the EKF (red) for essentially the entire flight, the shaded

TABLE I

Batch estimator on line 1003.02 (compensated stinger); horizontal DRMS [m]. Free-inertial reference $114.6\,\mathrm{m}$. The RTS and GN/QR solvers agree, confirming a common MAP estimate.

Segment	Estimator	DRMS
full line	EKF	28.3
full line	FGO (RTS)	14.3
full line	**FGO (RTS, Huber)**	**14.0**
$10\,\mathrm{min}$	EKF	47.9
$10\,\mathrm{min}$	FGO (RTS)	18.4
$10\,\mathrm{min}$	FGO (GN/QR)	19.3
$10\,\mathrm{min}$	FGO (GN/QR, Huber)	18.4

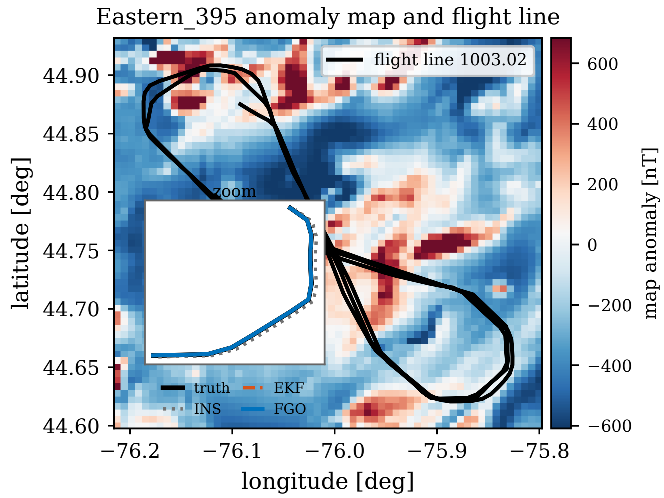

Fig. 4. Eastern anomaly map and flight line 1003.02. The inset zooms a mid-flight window where the free-inertial (INS) track drifts off the truth while the EKF and the factor-graph (FGO) estimates track it.

region marking the improvement. Fig. 6 gives the empirical error distribution: the FGO curve reaches its 95th percentile at less than half the error of the EKF, and both bound the free-inertial error by an order of magnitude. On a $10\,\mathrm{min}$ segment the iterated-RTS and sparse-GN/QR solvers agree to within $1\,\mathrm{m}$ (Table I), confirming that both numerically reach the same MAP estimate as predicted by Lemma 1. The Huber kernel changes the batch result by less than $0.3\,\mathrm{m}$ here, as expected for the already-compensated stinger sensor whose residuals are close to Gaussian; its value appears in the cold-start cases below, where transient outliers are large.

C. Cold-start line vs. online EKF+TL+NN

The central comparison is the cold start on uncompensated cabin magnetometers, where the interference is present and must be learned online. Table II and Fig. 7 report the primary evaluation line 1007.06 ($87\,\mathrm{min}$, full length) on Mag 4 and Mag 5. A static-TL batch forces a single coefficient vector over the whole line; it underfits the time-varying interference and degrades to $123.7/68.1\,\mathrm{m}$. Allowing the compensation to adapt with a $5\,\mathrm{min}$ window recovers it to $37.0/15.1\,\mathrm{m}$, and the Huber kernel further trims transient outliers to $32.6/14.2\,\mathrm{m}$. Our reproduced online EKF+TL+NN reaches $40.0/17.5\,\mathrm{m}$—inside the band published for that filter [23], confirming a

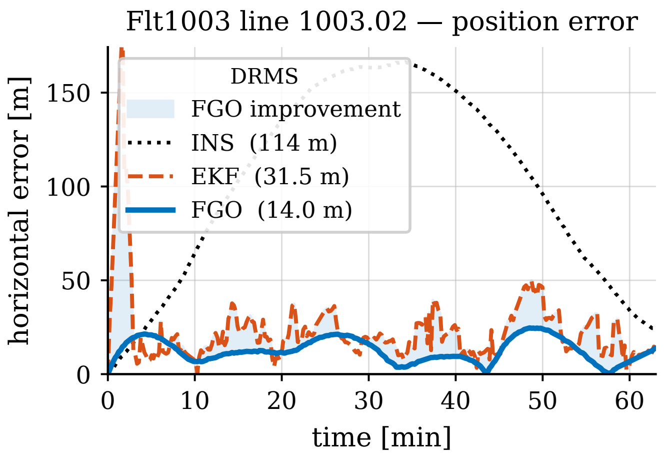

Fig. 5. Horizontal position error on line 1003.02. The batch factor graph (FGO, blue) holds a tighter bound than the EKF (red) across the line; the shaded band marks the FGO improvement, and free inertial (INS) diverges.

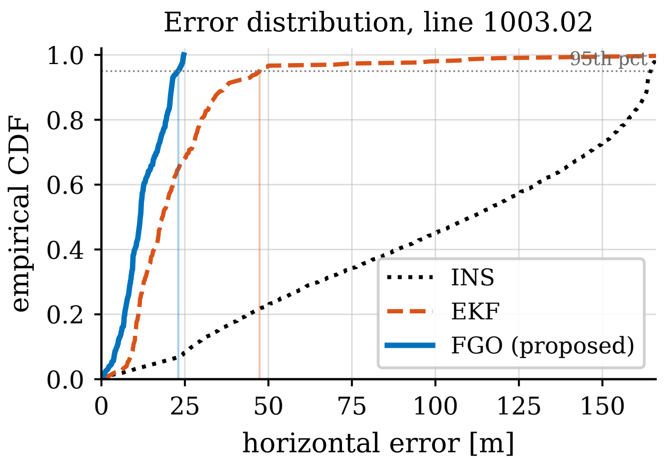

Fig. 6. Empirical CDF of horizontal error on line 1003.02. The proposed FGO reaches its 95th percentile at less than half the EKF's error; both bound the free-inertial error by an order of magnitude.

faithful reproduction—while the published TL-only EKF is weaker still. The window smoother therefore matches or beats the NN-augmented filter on both sensors *without any neural network*, using only the linear TL basis carried as graph variables. We attribute the margin to the window look-ahead: the coefficients are identified from data on both sides of each committed epoch, whereas the EKF must commit with past data only.

D. Breadth across lines, flights, and maps

Table III runs the same cold-start TL model through the causal online-TL EKF and the fixed-lag window smoother on five long survey lines spanning three flights and two maps. The window smoother is better on 8 of 9 cold-start cases. On the noisy Mag 4 the causal EKF diverges from the cold start to tens of kilometers on three lines, whereas the window smoother stays bounded, because its per-window relinearization lets later-observable calibration protect earlier navigation. The lone exception is a 14 min line on which every method is weak, reflecting a map-information floor for short lines.

TABLE II
COLD START, LINE 1007.06, UNCOMPENSATED CABIN MAGNETOMETERS; DRMS [M]. INS 318 m; EKF ON COMPENSATED MAG 1 $\approx$ 20 m.

Method (no NN unless noted)	Mag 4	Mag 5
FGO-online, static batch TL	123.7	68.1
FGO-online, window 5 min	37.0	15.1
FGO-online, window 5 min + Huber	**32.6**	**14.2**
EKF+TL+NN, reproduced [23]	40.0	17.5
EKF+TL+NN, published [23]	37	14
EKF+TL-only, published [23]	58	15

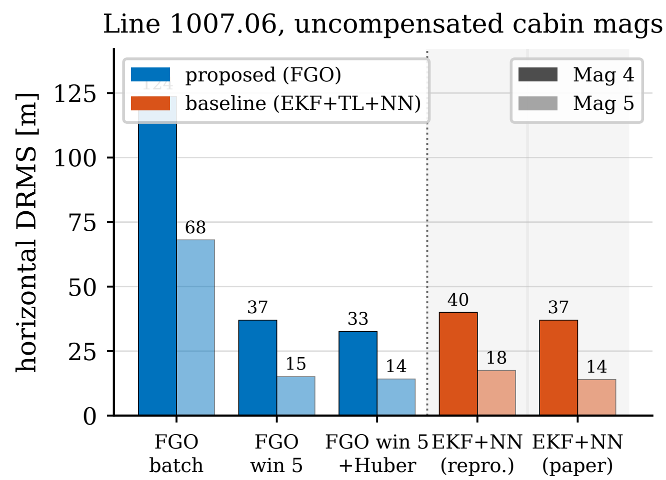

Fig. 7. Line 1007.06 cold start on the uncompensated cabin magnetometers. The NN-free fixed-lag window (left of the divider) matches or beats the reproduced and published EKF+TL+NN (shaded, right).

Reading the table by case sharpens the picture. On the three lines where the causal EKF diverges (1003.02, 1003.08, and 1007.02 on Mag 4), the divergence is to 1.7–42×10^3 m—the filter has committed to a wrong position from which the later, correct calibration cannot recover it—whereas the window smoother stays at 26 m to 43 m. On the well-conditioned Mag 5 sensor both estimators are stable and the window still wins on four of five lines, by 3 m to 17 m, the residual benefit of the look-ahead. The single case the EKF wins (1006.08 Mag 5, 117.5 vs 122.0 m) is the short line where neither method does well and the difference is within run-to-run variability. Averaging the bounded cases, the window reduces DRMS by 28 % relative to the causal EKF while removing the divergence mode entirely—the two effects the fixed-lag formulation was designed to provide.

E. Sensor-error factors (simulation)

To exercise the sensor-error factors of Section III we inject a known heading error (8 nT RMS), hard-iron bias, linear drift, and dead-zone noise into a simulated flight (free-inertial reference 31.1 m) and enable the corresponding graph variables cumulatively. Table IV reports the result. With no sensor-error states the estimate is *worse* than free inertial (36.6 m): the unmodeled error corrupts the map aiding, exactly the mechanism of Proposition 1. Enabling the factors drives the error down monotonically to 4.8 m, an 87 % reduction. The largest gains come from the hard-iron bias and the linear

TABLE III
BREADTH: COLD-START CABIN MAGNETOMETERS, DRMS [M]; CAUSAL ONLINE-TL EKF VS. 5 min-WINDOW SMOOTHER (HUBER). "DIV." DENOTES DIVERGENCE BEYOND 10 km.

Flight	Line	Map	Mag	EKF-online	FGO-win
Flt1003	1003.02	E	4	div.	**42.6**
Flt1003	1003.02	E	5	28.1	**21.7**
Flt1003	1003.08	R	4	div.	**26.1**
Flt1003	1003.08	R	5	21.1	**12.4**
Flt1006	1006.08	E	4	div.	**193.9**
Flt1006	1006.08	E	5	**117.5**	122.0
Flt1007	1007.02	E	4	div.	**38.6**
Flt1007	1007.02	E	5	31.6	**14.5**
Flt1007	1007.06	R	4	46.7	**32.7**
Flt1007	1007.06	R	5	17.8	**13.8**
FGO better					**8 / 9**

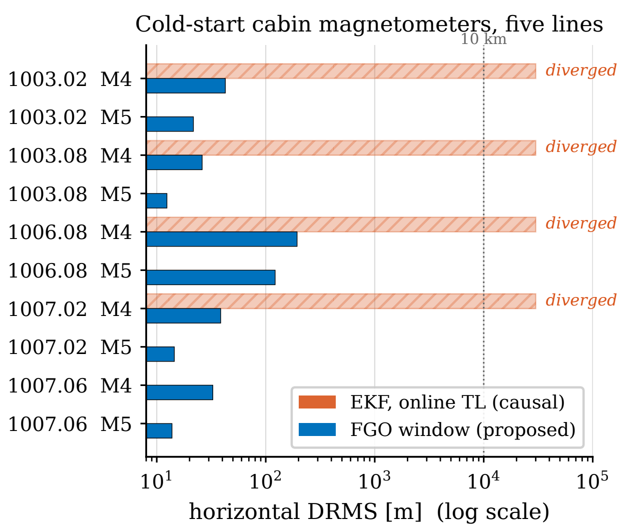

Fig. 8. Breadth study (Table III) on a log DRMS axis. Hatched bars mark causal EKF divergence beyond 10 km; the fixed-lag window (blue) stays bounded throughout.

drift, which the estimator recovers cleanly (drift $\hat{\gamma}_1 = 0.019$ versus a true 0.020). The heading harmonics help least: level flight sweeps the sensor–field angle by only about 12°, so the harmonic regressor is barely excited and its coefficients are poorly identified—an observability property of the trajectory, consistent with the Remark of Section IV, that a maneuvering flight would relieve. The Huber kernel gives essentially the same values (within 0.3 m), as expected for this near-Gaussian simulation.

F. Computational considerations

Each window is a sparse nonlinear least-squares problem of fixed size $O\big(L_w(n+m)\big)$; the block-bidiagonal-plus-shared-column Jacobian (Fig. 2) is factored in $O(L_w)$ by the square-root solver [30], and the number of Gauss–Newton iterations is small (typically 3–5) because the problem is mildly nonlinear away from observability collapse. The per-window cost is therefore constant in the line length and the overall complexity is linear in the number of epochs, matching a filter's asymptotic cost while retaining the look-ahead. Memory is bounded by the window, not the flight. The one-time overhead relative

TABLE IV
SENSOR-ERROR FACTORS (SIMULATION), ENABLED CUMULATIVELY; HORIZONTAL DRMS [M]. FREE-INERTIAL REFERENCE 31.1 m. VALUES FOR THE NON-ROBUST SOLVER; THE HUBER SOLVER AGREES WITHIN 0.3 m.

Graph variables enabled	DRMS
none (map aiding only)	36.6
+ heading harmonics (K=2)	19.1
+ dead-zone weight	19.2
+ hard-iron bias	8.4
+ **linear drift (full)**	**4.8**

TABLE V
ESTIMATOR FAMILIES FOR COLD-START MAP-MATCHING COMPENSATION. ✓ = YES, ✗ = NO, (∼) = LIMITED.

Estimator	Future data	Comp. state	Non-Gauss.	Bounded latency
Causal EKF [4]	✗	(∼)	✗	✓
Particle / grid filter [13]	✗	✗	✓	✓
Online EKF+TL+NN [23]	✗	✓	(∼)	✓
Batch FGO	✓	✓	✓	✗
Proposed (fixed-lag FGO)	✓	✓	✓	✓

to an EKF is the window factorization and the IRLS reweighting, which in our unoptimized research implementation runs comfortably faster than real time on the SGL sampling rate.

VII. DISCUSSION

A. Why the window helps

The advantage is not smoothing per se, which equals the RTS smoother [39] in the linear–Gaussian case, but the joint windowed estimation of a *time-varying* compensation together with navigation. A causal filter must commit each state online with no future context; under a large uncompensated cold-start residual it has only the position states with which to explain the mismatch, attributes it to position, and—as Table III shows on the noisy Mag 4—can diverge to tens of kilometres. The fixed-lag window defers commitment by L_o epochs, so the compensation is identified from data on both sides of each committed state before that state is fixed; later-observable calibration then protects earlier navigation. This is the same structural reason that batch and incremental smoothers outperform filtering in visual–inertial estimation [36], [38], here applied to the compensation problem. Table V positions the estimator families along the axes that matter for cold-start compensation.

B. The role of observability

Proposition 1 delimits when the joint problem is well-posed at all: the map-gradient and compensation subspaces must intersect only trivially. Its two corollaries organize the empirical behavior. Exogeneity (Corollary 1) is why excluding the scalar magnetometer from the network inputs was necessary for the baseline to converge, and why the attitude-only TL basis is safe at any dimension. The map-information floor of the Remark is why the 14 min line 1006.08 is hard for every estimator—there is simply little independent $\mathbf{g}_t$ to exploit—and it is the one case the window does not win. We stress that the condition is geometric and predictive only through these qualitative rules; our attempts at a scalar screening statistic did not generalize, which we report as a negative result rather than overclaim.

C. Deployment considerations

The estimator is compatible with real-time operation. Its committed output lags real time by the overlap L_o (here $2\,\mathrm{min}$), a latency that is acceptable for the map-aiding role—the INS provides the high-rate, zero-latency solution, and the smoother periodically corrects it—but that a designer trades against accuracy by choosing L_o. Because each window is bounded, both computation and memory are constant per commit and independent of flight duration, unlike a growing batch. The factor interface is modular: the compensation, sensor-error, and map factors are added or removed without touching the solver, so the same implementation covers the calibrated (frozen $\boldsymbol{\beta}$), cold-start (adaptive $\boldsymbol{\beta}$), and augmented-sensor cases. Integration with an operational system therefore amounts to feeding the INS mechanization, fluxgate, and scalar magnetometer streams into the graph and reading back the corrected position at the commit rate. The main tuning burden is the two window parameters and the process noise $\mathbf{Q}^{\beta}$, all of which have clear physical interpretations.

D. Limitations

Several limitations bound the present study, and we state them plainly. The real-data results are single-run DRMS on five lines; Monte-Carlo statistics over initializations and the full SGL line set remain to be run. We compare against a causal online-TL EKF and a reproduced online EKF+TL+NN, but not against a like-for-like grid or particle-filter map-matching baseline [10], [13], which would sharpen the estimator-structure claim. The sensor-error factors are validated in simulation only; a flight demonstration with genuine optically-pumped-magnetometer heading error is future work. Finally, short lines remain hard for every method, an information limit rather than an estimator one. None of these affects the two central, reproducible findings: the window smoother matches or beats the NN-augmented filter without a network, and it remains bounded where the causal filter diverges.

VIII. Conclusion

Casting online Tolles–Lawson aeromagnetic compensation as time-varying factor-graph variables in a fixed-lag sliding-window smoother yields a neural-network-free cold-start calibrator that, on real SGL 2020 data, matches or beats a reproduced online EKF+TL+NN on the primary line and generalizes across five lines, three flights, and two maps, remaining bounded where the causal EKF diverges. An exact observability condition and its exogeneity corollary explain the behavior and guide the choice of compensation features.

Appendix A
Pinson Error-State Dynamics

The navigation-error block of $\boldsymbol{\Phi}_t$ in (2) follows the Pinson model in a local north-east-down frame [2], [1]. With position, velocity, and attitude errors $(\delta\mathbf{p}, \delta\mathbf{v}, \boldsymbol{\psi})$ and sensor biases $(\mathbf{b}_a, \mathbf{b}_g)$, the continuous error dynamics are

$$\begin{aligned} \dot{\delta\mathbf{p}} &= \mathbf{F}_{pp}\,\delta\mathbf{p} + \delta\mathbf{v}, \\ \dot{\delta\mathbf{v}} &= \mathbf{F}_{vp}\,\delta\mathbf{p} + \mathbf{F}_{vv}\,\delta\mathbf{v} - [\mathbf{f}^n\times]\,\boldsymbol{\psi} + \mathbf{C}_b^n\mathbf{b}_a, \\ \dot{\boldsymbol{\psi}} &= \mathbf{F}_{\psi p}\,\delta\mathbf{p} + \mathbf{F}_{\psi v}\,\delta\mathbf{v} - [\boldsymbol{\omega}_{in}^n\times]\,\boldsymbol{\psi} - \mathbf{C}_b^n\mathbf{b}_g, \end{aligned} \tag{12}$$

where $\mathbf{f}^n$ is the specific force, $\mathbf{C}_b^n$ the body-to-nav rotation, $\boldsymbol{\omega}_{in}^n$ the inertial rate, $[\cdot\times]$ the skew operator, and the $\mathbf{F}_{\bullet\bullet}$ blocks collect the transport-rate, gravity-gradient, and Coriolis terms. The biases follow first-order Gauss–Markov or random-walk models, as does the scalar magnetic disturbance state S with correlation time τ. The discrete $\boldsymbol{\Phi}_t = \exp(\mathbf{F}_t\,\Delta t)$ and $\mathbf{Q}_t$ are obtained by standard van Loan discretization [40]. Only $\mathbf{g}_t^\top$, the position row of the measurement Jacobian (9), couples this block to the map.

Appendix B
Tolles–Lawson Basis

The basis row $\mathbf{A}_t$ of (6) is built from the fluxgate direction cosines $\hat{\mathbf{b}} = [\,u\;\;v\;\;w\,]^\top$ (with $u^2+v^2+w^2=1$) and the scalar field magnitude B [6], [18]. The permanent moment contributes the three cosines (u, v, w); the induced magnetization, symmetric in the field, contributes the six products $B\,(u^2, uv, uw, v^2, vw, w^2)$; and the eddy-current field, proportional to the rate of change of the cosines, contributes the nine products $B\,(u\dot{u}, u\dot{v}, \ldots, w\dot{w})$. Stacking these gives an eighteen-term basis, optionally augmented by a constant bias, and $c_t = \mathbf{A}_t^\top\boldsymbol{\beta}_t$ is linear in the coefficients $\boldsymbol{\beta}_t$—the property that lets them enter the factor graph as ordinary linear-measurement variables. In the cold-start regime the calibration flight that classically fixes $\boldsymbol{\beta}$ is unavailable, which is precisely why we carry it as a graph variable.

Acknowledgment

The experiments use the publicly released SGL 2020 challenge dataset [5].

References

[1] P. D. Groves, *Principles of GNSS, Inertial, and Multisensor Integrated Navigation Systems*, 2nd ed. Boston, MA: Artech House, 2013.
[2] D. H. Titterton and J. L. Weston, *Strapdown Inertial Navigation Technology*, 2nd ed. Institution of Engineering and Technology, 2004.
[3] A. Canciani and J. Raquet, "Absolute positioning using the Earth's magnetic anomaly field," *NAVIGATION: Journal of the Institute of Navigation*, vol. 63, no. 2, pp. 111–126, 2016.
[4] ——, "Airborne magnetic anomaly navigation," *IEEE Transactions on Aerospace and Electronic Systems*, vol. 53, no. 1, pp. 67–80, 2017.
[5] A. R. Gnadt, J. Belarge, A. Canciani, L. Conger, J. Curro, A. Edelman, P. Morales, A. P. Nielsen, M. Ryan, and A. Wollaber, "Signal enhancement for magnetic navigation challenge problem," *arXiv preprint*, 2020.
[6] W. E. Tolles and J. D. Lawson, "Magnetic compensation of MH-1A aircraft," Airborne Instruments Laboratory, Tech. Rep. Report 1, 1950.
[7] R. E. Kalman, "A new approach to linear filtering and prediction problems," *Journal of Basic Engineering*, vol. 82, no. 1, pp. 35–45, 1960.
[8] J. P. Golden, "Terrain contour matching (TERCOM): A cruise missile guidance aid," in *Image Processing for Missile Guidance, Proc. SPIE*, vol. 238, 1980, pp. 10–18.
[9] L. Hostetler and R. Andreas, "Nonlinear Kalman filtering techniques for terrain-aided navigation," *IEEE Transactions on Automatic Control*, vol. 28, no. 3, pp. 315–323, 1983.

[10] N. Bergman, "Recursive Bayesian estimation: Navigation and tracking applications," Ph.D. dissertation, Linköping University, Linköping, Sweden, 1999.

[11] F. Gustafsson, F. Gunnarsson, N. Bergman, U. Forssell, J. Jansson, R. Karlsson, and P.-J. Nordlund, "Particle filters for positioning, navigation, and tracking," *IEEE Transactions on Signal Processing*, vol. 50, no. 2, pp. 425–437, 2002.

[12] K. B. Anonsen and O. Hallingstad, "Terrain aided underwater navigation using point mass and particle filters," in *IEEE/ION Position, Location, and Navigation Symposium (PLANS)*, 2006, pp. 1027–1035.

[13] P.-J. Nordlund and F. Gustafsson, "Marginalized particle filter for accurate and reliable terrain-aided navigation," *IEEE Transactions on Aerospace and Electronic Systems*, vol. 45, no. 4, pp. 1385–1399, 2009.

[14] A. J. Canciani, "Magnetic navigation on an F-16 aircraft using online calibration," *IEEE Transactions on Aerospace and Electronic Systems*, vol. 58, no. 1, pp. 420–434, 2022.

[15] S. J. Julier and J. K. Uhlmann, "Unscented filtering and nonlinear estimation," *Proceedings of the IEEE*, vol. 92, no. 3, pp. 401–422, 2004.

[16] A. Barrau and S. Bonnabel, "The invariant extended Kalman filter as a stable observer," *IEEE Transactions on Automatic Control*, vol. 62, no. 4, pp. 1797–1812, 2017.

[17] Y. Bar-Shalom, X.-R. Li, and T. Kirubarajan, *Estimation with Applications to Tracking and Navigation*. John Wiley & Sons, 2001.

[18] P. Leliak, "Identification and evaluation of magnetic-field sources of magnetic airborne detector equipped aircraft," *IRE Transactions on Aerospace and Navigational Electronics*, vol. ANE-8, no. 3, pp. 95–105, 1961.

[19] S. H. Bickel, "Small signal compensation of magnetic fields resulting from aircraft maneuvers," *IEEE Transactions on Aerospace and Electronic Systems*, vol. AES-15, no. 4, pp. 518–525, 1979.

[20] P. M. Williams, "Aeromagnetic compensation using neural networks," *Neural Computing & Applications*, vol. 1, no. 3, pp. 207–214, 1993.

[21] G. Noriega, "Aeromagnetic compensation in gradiometry—performance, model stability, and robustness," *IEEE Geoscience and Remote Sensing Letters*, vol. 12, no. 1, pp. 117–121, 2015.

[22] A. R. Gnadt, "Machine learning-enhanced magnetic calibration for airborne magnetic anomaly navigation," Ph.D. dissertation, Massachusetts Institute of Technology, Cambridge, MA, 2022.

[23] Hager *et al.*, "Airborne magnetic navigation with neural-network-augmented online calibration," *arXiv preprint*, 2026, citation metadata to be finalized against the published version.

[24] F. R. Kschischang, B. J. Frey, and H.-A. Loeliger, "Factor graphs and the sum-product algorithm," *IEEE Transactions on Information Theory*, vol. 47, no. 2, pp. 498–519, 2001.

[25] F. Dellaert and M. Kaess, "Factor graphs for robot perception," *Foundations and Trends in Robotics*, vol. 6, no. 1-2, pp. 1–139, 2017.

[26] F. Dellaert, "Factor graphs and GTSAM: A hands-on introduction," Georgia Institute of Technology, Tech. Rep. GT-RIM-CP&R-2012-002, 2012.

[27] B. Triggs, P. F. McLauchlan, R. I. Hartley, and A. W. Fitzgibbon, "Bundle adjustment—a modern synthesis," in *Vision Algorithms: Theory and Practice*, 2000, pp. 298–372.

[28] G. Grisetti, R. Kümmerle, C. Stachniss, and W. Burgard, "A tutorial on graph-based SLAM," *IEEE Intelligent Transportation Systems Magazine*, vol. 2, no. 4, pp. 31–43, 2010.

[29] R. Kümmerle, G. Grisetti, H. Strasdat, K. Konolige, and W. Burgard, "g2o: A general framework for graph optimization," in *IEEE International Conference on Robotics and Automation (ICRA)*, 2011, pp. 3607–3613.

[30] F. Dellaert and M. Kaess, "Square root SAM: Simultaneous localization and mapping via square root information smoothing," *The International Journal of Robotics Research*, vol. 25, no. 12, pp. 1181–1203, 2006.

[31] M. Kaess, A. Ranganathan, and F. Dellaert, "iSAM: Incremental smoothing and mapping," *IEEE Transactions on Robotics*, vol. 24, no. 6, pp. 1365–1378, 2008.

[32] M. Kaess, H. Johannsson, R. Roberts, V. Ila, J. J. Leonard, and F. Dellaert, "iSAM2: Incremental smoothing and mapping using the Bayes tree," *The International Journal of Robotics Research*, vol. 31, no. 2, pp. 216–235, 2012.

[33] R. M. Eustice, H. Singh, and J. J. Leonard, "Exactly sparse delayed-state filters for view-based SLAM," *IEEE Transactions on Robotics*, vol. 22, no. 6, pp. 1100–1114, 2006.

[34] G. Sibley, L. Matthies, and G. Sukhatme, "Sliding window filter with application to planetary landing," *Journal of Field Robotics*, vol. 27, no. 5, pp. 587–608, 2010.

[35] S. Leutenegger, S. Lynen, M. Bosse, R. Siegwart, and P. Furgale, "Keyframe-based visual–inertial odometry using nonlinear optimization," *The International Journal of Robotics Research*, vol. 34, no. 3, pp. 314–334, 2015.

[36] C. Forster, L. Carlone, F. Dellaert, and D. Scaramuzza, "On-manifold preintegration for real-time visual–inertial odometry," *IEEE Transactions on Robotics*, vol. 33, no. 1, pp. 1–21, 2017.

[37] V. Indelman, S. Williams, M. Kaess, and F. Dellaert, "Information fusion in navigation systems via factor graph based incremental smoothing," *Robotics and Autonomous Systems*, vol. 61, no. 8, pp. 721–738, 2013.

[38] T. Qin, P. Li, and S. Shen, "VINS-Mono: A robust and versatile monocular visual-inertial state estimator," *IEEE Transactions on Robotics*, vol. 34, no. 4, pp. 1004–1020, 2018.

[39] H. E. Rauch, F. Tung, and C. T. Striebel, "Maximum likelihood estimates of linear dynamic systems," *AIAA Journal*, vol. 3, no. 8, pp. 1445–1450, 1965.

[40] J. A. Farrell, *Aided Navigation: GPS with High Rate Sensors*. McGraw-Hill, 2008.

[41] P. J. Huber, "Robust estimation of a location parameter," *The Annals of Mathematical Statistics*, vol. 35, no. 1, pp. 73–101, 1964.

[42] N. Sünderhauf and P. Protzel, "Switchable constraints for robust pose graph SLAM," in *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2012, pp. 1879–1884.

[43] P. Agarwal, G. D. Tipaldi, L. Spinello, C. Stachniss, and W. Burgard, "Robust map optimization using dynamic covariance scaling," in *IEEE International Conference on Robotics and Automation (ICRA)*, 2013, pp. 62–69.